1 Social Optimum: Definition

Social cost. An edge e crossed by f agents in the same timestep costs each of them $c_e(f) = a_e + b_e f$. The social cost of a run is the total paid by all agents,

$$SC = \sum_t \sum_e f_e(t)\, c_e(f_e(t)),$$

where $f_e(t)$ is the number of agents crossing e in step t.

Social optimum. SC_{opt} is the lowest social cost any routing of the N agents can achieve under the simulator's movement rules:

1. all agents start at $t = 0$;
2. every agent that has not arrived crosses exactly one edge per step (no waiting);
3. no agent moves back towards its origin or revisits a node (`forward_only`).

Agents are indivisible. Edge costs do not change with N, so SC_{opt} is computed separately for each population size and reported per agent, SC_{opt}/N.

As a choice of paths. Under these rules an agent crosses the k-th edge of its path at step $k-1$. A routing is therefore just a path for each agent, and

$$SC_{\text{opt}} = \min_{\text{paths}} \sum_{e,\,t} X_{e,t}\,(a_e + b_e X_{e,t}),$$

where $X_{e,t}$ is the number of agents whose $(t+1)$-th edge is e. Agents congest each other only if they cross the same edge in the same step.

Relation to the static optimum. The classical static optimum is $\min \sum_e x_e\, c_e(x_e)$ over the same paths, where x_e is the total number of agents using e: every user of an edge is charged for all of its users, whenever they cross.

- If every edge is always crossed at the same step (Pigou, Diamond, Grid), then $X_{e,t} = x_e$ and the two optima are equal.
- Otherwise SC_{opt} is never higher, since $\sum_t X_{e,t}^2 \le \left(\sum_t X_{e,t}\right)^2$: spreading an edge's users over several steps can only reduce congestion. For Braess and Sioux Falls it is strictly lower.

Every simulated run costs at least SC_{opt}, so $SC/SC_{\text{opt}} \ge 1$ measures how far a run is from optimal.